Droughts and Local Livestock Market Conditions

Abstract

Drought affects cattle marketing decisions through culling the breeding herd and early weaning calves, but less is known about how these adjustments appear in local livestock auctions. We combine United States Department of Agriculture auction reports with pasture-weighted drought conditions surrounding cattle auction barns to estimate how drought in each quarter of the year before a sale affects the reported number of head sold, feeder cattle average weights, and prices for feeder cattle and cull cows. Severe drought increases the number of cattle marketed, with the largest effects for cull cows and feeder heifers, consistent with culling the breeding herd and delaying heifer retention. Steers and heifers are also sold at lighter weights, consistent with drought affecting weaning and marketing decisions. Price discounts are relatively small and concentrated in the first three months after drought, especially for lighter feeder cattle, while positive effects appear at longer lags for heavier categories. Together, these results show how drought affects local livestock markets by changing the number, weight, and composition of cattle marketed, with smaller effects on prices. Findings have broader implications for understanding the impacts of producer decision making on the U.S. cattle cycle.

Record-high beef prices have generated considerable interest among policymakers in increasing beef supplies and lowering prices for U.S. consumers. Beef and veal prices increased 3.5% from September 2025 to September 2026 and nearly 17.8% from September 2024 to September 2026 (U.S. Bureau of Labor Statistics 2026). In response to high beef prices, several policies have been implemented to increase cattle and beef supplies. These include efforts to reopen the southern border to cattle imports from Mexico following closures related to New World screwworm, additional tariff-free beef imports from Argentina, and a temporary 300,000 metric ton increase in the tariff-rate quota for lean beef trimmings from eligible countries (U.S. Department of Agriculture, Animal and Plant Health Inspection Service 2026; The White House 2026a,b). At the same time, USDA has introduced policies aimed at rebuilding the U.S. cattle herd and encouraging investments in cattle production (U.S. Department of Agriculture 2026). The challenge is that increasing cattle supplies ultimately depends on decisions made by individual cattle producers. Producers must decide whether to market animals today or retain breeding animals that can increase cattle supplies in future years. These decisions are affected by current and expected profitability, but also by production conditions that can alter the costs of retaining cattle. When aggregated across producers and over time, these individual production and marketing decisions determine changes in the U.S. cattle inventory and contribute to the broader cattle cycle.

The cattle cycle is the most defining characteristic of the U.S. beef cattle industry. It describes cyclical changes in total U.S. cattle inventories driven by delayed responses to changing profitability because of the biology inherent to cattle production (Tonsor and Mitchell 2017). The economic dynamics of cattle cycles, which last on average between 9 and 13 years, are well understood in the literature (e.g., Rosen, Murphy, and Scheinkman 1994; Schmitz 1997; Marsh 1999; Hamilton and Kastens 2000). Each cattle cycle has its own set of economic circumstances, but there are long-run trends across cycles, including the observation that cycles are becoming less pronounced, coinciding with increased beef production per cow (Tonsor and Mitchell 2017; Tonsor, Luke, and Anderson 2022). Understanding cattle cycles and the factors affecting them is important for strategic planning at both the farm and industry levels.

Beyond the relationship between profitability and cattle inventories, more recent research has emphasized the exacerbating effects of drought on cattle numbers and production (Leister, Paarlberg, and Lee 2015). These studies have focused on cattle inventory and price responses to drought conditions using spatially and temporally aggregated data but provide valuable insights into the longer-run effects of drought. One such insight is the persistence of drought shocks on breeding inventories beyond the drought period because of the time required for herd rebuilding.

Common drought response strategies for cattle producers include supplemental feeding, early weaning of calves, and culling cows. The extent to which producers use these strategies depends on the severity and duration of drought, but collectively these decisions can have important implications for cattle inventories and markets. In particular, increased culling and other drought-induced sales can increase the number of cattle marketed in the short run while reducing breeding inventories and cattle supplies in later years. These adjustments can have both local and broader market effects, with increased cattle sales potentially affecting prices at the time of liquidation and smaller inventories affecting prices as the industry rebuilds. Yet, relatively little is known about how drought-induced management decisions translate into changes in cattle marketings and prices at the individual sale level, or how these effects evolve over the course of a drought.

In this paper, we determine the impacts of drought on cattle auction prices and sale volumes for feeder cattle and cull cows across markets, weeks, and feeder weight classes. To do this, we leverage cattle auction data from the U.S. Department of Agriculture's Agricultural Marketing Service to construct price series for homogeneous groups of cattle as well as the reported number of head and average weight for each group. We then construct sale-specific measures of drought by matching each auction and sale date to pasture-weighted drought conditions within 100 miles of the auction using the Standardized Precipitation Evapotranspiration Index (SPEI) from gridMET (Abatzoglou 2013). Drought conditions are measured over four three-month intervals covering the 12 months preceding each sale.

From our regression results, we find three main results. First, drought increases the number of cattle marketed, with the largest effects for cull cows and heifers. Second, feeder cattle are marketed

at lighter weights during drought, consistent with producers adjusting weaning and market timing. Third, drought negatively impacts auction prices, but these price effects are relatively small and concentrated in the months immediately following drought. Together, these results show how producer response to drought impact cattle sales and short-run prices at local auction markets.

This paper contributes to two related areas of the literature. First, it extends the literature examining the effects of drought on cattle inventories and markets by estimating how local drought conditions affect cattle marketings and prices at individual cattle auctions. Previous studies have largely examined these relationships using national or state-level cattle inventories and annual or quarterly prices (Leister, Paarlberg, and Lee 2015). Second, this paper contributes to the literature examining factors affecting cattle prices. Most studies in this literature focus on the effects of physical cattle characteristics on price (Schroeder et al. 1988; Zimmerman et al. 2012; Mitchell, Peel, and Brorsen 2018; Martinez, Boyer, and Burdine 2021). Mallory et al. (2016) do not directly estimate the relationship between drought and cattle prices but find a relationship between wheat pasture acreage and feeder cattle basis in Oklahoma. Lanfranco and Castaño-Sánchez (2017) include soil moisture and pasture conditions in their hedonic model of cattle prices in Uruguay and find a negative relationship between prices and improving pasture conditions. We extend this literature by directly estimating how local drought conditions affect cattle marketings and prices for both feeder cattle and cull cows and how these effects evolve over the course of a drought.

Theoretical Model

U.S. beef cattle production involves several stages, including cow-calf and finishing. In the cow-calf stage, beef cows are kept as breeding stock to produce and raise calves. After weaning, calves can proceed through the subsequent stages of the U.S. beef supply chain. A share of the female calves are retained as replacements for the breeding herd and will contribute to calf production over the following years. Because of this characteristic, beef cows can be treated as either capital goods,

when retained for breeding, or consumption goods, when sold for slaughter (Jarvis 1974). Current culling and breeding-stock decisions therefore affect future cattle inventories through biological constraints and production processes (Foster and Burt 1992; Rosen, Murphy, and Scheinkman 1994). Producers' decisions depend on expected future prices (Nerlove and Fornari 1998; Aadland and Bailey 2001), feed costs (Ramsey et al. 2005), and weather conditions (Leister, Paarlberg, and Lee 2015).

Our conceptual approach consists of explaining how cow-calf producers adjust their weekly livestock-marketing decisions in response to drought exposure. We assume that, at the beginning of each week, the rancher decides on the number of animals to sell, the number of calves to retain for at least one additional week to gain additional weight, the number of heifers to retain as breeding replacements, and the amount of supplemental feed to purchase. This weekly timing is consistent with the very short-run nature of livestock-marketing decisions because livestock are not a storeable commodity.

Rancher's Optimization Problem

Consider a rancher raising breeding cows and feeder cattle whose objective is to maximize the expected present value of profits. At the beginning of week t, the rancher observes K_t, the breeding-cow inventory; N_{it}, the number of available feeder cattle of sex i, with $i \in \{s, h\}$, where s denotes steers and h denotes heifers; W_{jt}, the average weight of cattle category j, with $j \in \{c, s, h\}$, where c denotes cows; F_t, the available forage; and Z_t, a vector containing prices, animal quality, and other production conditions. Feeder cattle may gain additional weight through continued retention, supported by maternal milk before weaning and by forage and supplemental feed. In contrast, breeding-cow weight is assumed to be maintained at a level necessary to preserve their health and reproductive potential.

The rancher decides on C_t, the number of cows to be culled; V_{it}, the number of feeder cattle of sex i to be sold; G_{it}, the number of feeder cattle of sex i to be retained for at least one additional week

for growth; R_t, the number of heifers to be retained as breeding replacements; and x_t, the quantity of supplemental feed to be purchased. These decisions are assumed to occur at the beginning of the week. The animals retained after these decisions consume forage and supplemental feed during the week. The rancher's decisions are constrained by the number of animals available in each category. Specifically, $0 \leq C_t \leq K_t$, and available feeder cattle must be allocated between current sale and retention. For steers, $V_{st} + G_{st} = N_{st}$, while heifers may be sold, retained for additional growth, or retained as breeding replacements: $V_{ht} + G_{ht} + R_t = N_{ht}$. All decision variables are nonnegative.

The ranchers optimization problem can be written as:

$$(1) \qquad V(\Omega_t) = \max_{C_t, V_{it}, G_{it}, R_t, x_t} \left\{ \pi_t(C_t, V_{it}, G_{it}, R_t, x_t; F_t) \quad + \beta E_t \left[V(\Omega_{t+1}) \right] \right\}.$$

subject to

$$(2) \qquad K_{t+1} = (1 - \delta_K)(K_t - C_t) + \rho R_{t-\ell_R},$$

and the feeder-cattle inventory constraint

$$(3) \qquad (1 - \mu_i) G_{i,t-1} + \phi_i K_{t-\ell_G} = V_{it} + G_{it} + \mathbf{1}\{i = h\} R_t, \qquad i \in \{s, h\}.$$

Equation (1) presents the rancher's dynamic optimization problem, where $\Omega_t = \left(K_t, N_{it}, W_{jt}, F_t, Z_t\right)$ is the vector of state variables. The variable π_t denotes current weekly profit, which includes revenues from cow and feeder-cattle sales and the costs of maintaining retained animals. For each animal category $i \in \{K, s, h, R\}$, maintenance costs are represented by $C_i\left(n_{it}, F_t, x_t, Z_t\right)$, which is a function of the number of retained animals, n_{it}; available forage, F_t; the quantity of supplemental feed purchased, x_t; and other production conditions, Z_t. Maintenance costs increase

with the number of retained animals and the quantity of supplemental feed purchased, such that $\partial C_i/\partial n_{it} > 0$ and $\partial C_i/\partial x_t > 0$. Cow-calf operations meet their feed needs primarily through grazing, greater forage availability reduces maintenance costs, such that $\partial C_i/\partial F_t < 0$. Finally, $0 < \beta < 1$ is the weekly discount factor, and $E_t\left[V(\Omega_{t+1})\right]$ is the expected value in the following week.

Equation (2) presents the breeding-herd transition constraint, where δ_K represents mortality and other nonmarket removals, excluding observed commercial culling. The parameters ρ and ℓ_R represent, respectively, the proportion of retained replacement heifers that successfully enter the breeding herd and the number of weeks required for them to do so.

Equation (3) describes the allocation and evolution of feeder-cattle inventories. The current numbers of steers and heifers in the feeder inventory depend on the numbers retained in the previous week, where μ_i represents mortality and other nonmarket removals. The number of cattle entering feeder category i is assumed to be proportional, ϕ_i, to the total number of animal kept as stock capital. The parameter ℓ_G is measured in weeks and represents gestation and the time required for calves to reach the feeder stage. Available feeder cattle are either sold during week t or retained for at least one additional week. Heifers may also be retained as breeding replacements. Thus, a reduction in feeder retention increases current sales.

For each cattle category, the rancher compares the animal's current market value, $P_{jt}(W_{jt})W_{jt}$, with the expected net value of retaining it (Perrin 1972). Here, $P_{jt}(W_{jt})$ is the price received per unit of weight for an animal in category j, and W_{jt} is its current weight. Although the individual rancher takes this price as given, local market prices may respond to aggregate cattle marketings. The optimal sale condition for each cattle category is represented by:

$$P_{jt}(W_{jt})W_{jt} \geq \begin{cases} -MC^G_{st} + \beta E_t\left[P_{s,t+1}(W_{s,t+1})W_{s,t+1}\right], & j = s, \quad (4) \\ \max\left\{\begin{array}{l} -MC^G_{ht} + \beta E_t\left[P_{h,t+1}(W_{h,t+1})W_{h,t+1}\right], \\ -MC^R_{ht} + \beta^{\ell_R} E_t\left[MV^B_{h,t+\ell_R}\right] \end{array}\right\}, & j = h, \quad (5) \\ -MC^{\text{saved}}_{ct} + \beta E_t\left[MV^B_{c,t+1}\right], & j = c. \quad (6) \end{cases}$$

For steers ($j = s$), the rancher compares the current sale value with the expected net value of retaining the animal for at least one additional week for further weight gain. The term MC^G_{st} represents the marginal cost of maintaining the steer during the additional week. The expected sale value in the following week depends on the steer's future weight, $W_{s,t+1}$, and the corresponding expected price per unit of weight, $P_{s,t+1}(W_{s,t+1})$. The steer is sold when its current market value is at least as large as its expected net retention value.

For heifers ($j = h$), the rancher allocates the currently available heifer inventory among three alternatives: current sale, temporary retention for additional growth, and retention as breeding replacements. The decision to retain a heifer for additional growth is similar to the steer-retention decision and is represented by the first expression in Equation (5). The term MC^G_{ht} represents the marginal cost of maintaining the heifer during the additional growth period. The second expression represents the expected net value of retaining the heifer as a breeding replacement. The term MC^R_{ht} represents the present value of the marginal costs incurred until the replacement heifer enters the breeding herd, while $MV^B_{h,t+\ell_R}$ represents her expected breeding value at that time. This breeding value includes the expected value of the calves produced over her productive life and her expected eventual market value when she is culled from the breeding herd. The rancher sells the heifer when her current market value is at least as large as both retention values.

For breeding cows ($j = c$), the rancher compares the current sale value with the expected value of retaining the cow in the breeding herd. The term $MV^B_{c,t+1}$ represents the cow's expected future

breeding value, including the value of future calf production and her expected future sale value. The term MC_{ct}^{saved} represents the marginal maintenance cost avoided by culling the cow.

Drought affects producers' decisions through both cost and production channels. Effective weekly forage availability, $F_t(\overline{F}_t, D_t)$, depends on the quantity and nutritional value of forage available under normal conditions, $\overline{F}_t$, and drought severity, D_t, with higher values of D_t indicating more severe drought. Effective forage availability decreases with drought severity (Porensky et al. 2026), such that $\partial F_t / \partial D_t < 0$. As forage becomes scarcer because of drought, producers rely more heavily on supplemental feed, such as hay or silage, thereby increasing feeding costs. Let x_t^* denote the optimal quantity of supplemental feed. Because the use of supplemental feed increases with drought severity, $\partial x_t^* / \partial D_t \geq 0$. The total effect of drought on maintenance costs is

$$\frac{dC_i}{dD_t} = \frac{\partial C_i}{\partial F_t}\frac{\partial F_t}{\partial D_t} + \frac{\partial C_i}{\partial x_t}\frac{\partial x_t^*}{\partial D_t} > 0, \qquad i \in \{K, s, h, R\}. \tag{7}$$

The first term, $(\partial C_i / \partial F_t)\,(\partial F_t / \partial D_t)$, captures the increase in maintenance costs caused by lower forage availability. The second term, $(\partial C_i / \partial x_t)\left(\partial x_t^* / \partial D_t\right)$, captures the additional expenditure on supplemental feed required to replace the reduction in forage. Thus, even if supplemental feeding is sufficient to maintain animal performance, drought raises the marginal cost of retention.

Reductions in forage quantity and quality directly limit the nutrients available to feeder cattle and may also reduce milk production among breeding cows (Scasta et al. 2015). For calves that have not yet been weaned, lower milk production reduces their primary source of nutrition and may further limit weight gain. Therefore, when supplemental feeding does not fully offset these adverse effects, drought reduces the future weight of feeder cattle:

$$\frac{\partial W_{i,t+1}}{\partial D_t} < 0, \quad i \in \{s, h\}.$$

Producers' decisions are affected by drought because it increases maintenance costs and reduces expected weight gain. Higher retention costs reduces the incentive to retain steers and heifers for

additional growth, or to retain heifers as breeding replacements, and increases incentive to cow culling. This leads to the first hypothesis:

Hypothesis 1: More severe drought increases the number of animals marketed.

Drought also affects the weight at which feeder cattle are marketed. Lower forage availability reduces expected weight gain during an additional week of retention. At the same time, higher maintenance costs shorten the optimal retention period by encouraging producers to sell animals earlier. This leads to the second hypothesis:

Hypothesis 2: More severe drought reduces the expected sale weight of feeder cattle and shift the composition of sales toward lighter weight categories.

Finally, the individual rancher takes auction prices as given, but the aggregation of ranchers' decisions can affect prices in local markets. When drought induces many producers in the same area to sell cattle, the resulting increase in local marketings places downward pressure on auction prices. This leads to the third hypothesis:

Hypothesis 3: More severe drought reduces local auction prices by increasing the number of cattle marketed within the affected area.

We examine these three hypotheses empirically. The following sections present the data and empirical models used in the analysis.

Data

The period of analysis is January 2000 to September 2026. Sales data come from USDA Agricultural Marketing Service (AMS) Market News, which reports sales at individual auction barns. We combine records obtained through the My Market News API (MARS), available in our sample beginning in 2018 (U.S. Department of Agriculture, Agricultural Marketing Service 2018), with historical Legacy Livestock Auction files covering 2000–2019 (U.S. Department of Agriculture,

Agricultural Marketing Service n.d.). The sources overlap in 2018–2019. For overlapping observations, we prioritize available MARS values and use Legacy values where the corresponding MARS information is unavailable, rather than adding observations from both sources together. We also use geocoding data from Census TIGER place centroids, and GNIS for unincorporated places. Coordinates represent the centroid of the barn's town or the location of its populated place. Video and other excluded nonregular sales are not used in the models. Data on sales include prices, weight, and the reported number of head sold in each cattle category, defined by sex, frame and muscle grade, and weight range. These are aggregated AMS detail rows, not individual transactions or total auction receipts. The sample covers 348 distinct auction barns, with coverage varying across outcomes and years because of differences in reporting availability.

We measure drought using the 90-day Standardized Precipitation Evapotranspiration Index (SPEI-90) from the gridMET drought dataset (Abatzoglou 2013, 2026). These drought indices are computed from gridMET, the 4 km daily weather dataset of the Climatology Lab. SPEI-90 is calculated for each 4 km grid cell every five days. For a given cell and date, it takes the water balance of the window (precipitation minus reference evapotranspiration, summed over the past 90 days) and compares it with the distribution for that cell and time of the year over the 1980–2016 reference period. The index is expressed on a standardized scale: negative values indicate drier conditions, positive values indicate wetter conditions, and zero represents typical conditions for that place and time of year. SPEI is one of the indicators considered by the US Drought Monitor (National Drought Mitigation Center 2026). Table 1 shows the U.S. Drought Monitor categories and corresponding SPEI thresholds.

Table 1. U.S. Drought Monitor Categories and Corresponding SPEI Thresholds

Category	Description	SPEI range	Percentile
None	Normal or wet	−0.49 or above	above 30
D0	Abnormally dry	−0.5 to −0.79	21 to 30
D1	Moderate drought	−0.8 to −1.29	11 to 20
D2	Severe drought	−1.3 to −1.59	6 to 10
D3	Extreme drought	−1.6 to −1.99	3 to 5
D4	Exceptional drought	−2.0 or less	0 to 2

Source: National Drought Mitigation Center, US Drought Monitor, Drought Classification. The US Drought Monitor considers several indicators. These ranges are the SPEI crosswalk. D2 can trigger the Livestock Forage Disaster Program when the applicable duration and grazing-period requirements are met.

After extracting the SPEI data, we use pastures as weight, and attribute to the nearest barn (within 100 miles) the pasture-weighted SPEI of the months before the sale. We identify pasture using the 2008–2022 editions of the USDA National Agricultural Statistics Service (NASS) Cropland Data Layer (Boryan et al. 2011; USDA National Agricultural Statistics Service 2008–2022). A 30 m pixel is pasture if it was classified as grassland/pasture or non-alfalfa hay in at least half of the years 2008–2022. Each 4 km grid cell with pasture is assigned to the nearest barn within 100 miles, and the SPEI of a barn on a sale date is the average of its cells weighted by their pasture pixels, using the most recent five-day value on or before the sale:

$$SPEI_{i,t}^{(90)} = \frac{\sum_{g \in G(i)} A_g \, SPEI_{g,t}^{(90)}}{\sum_{g \in G(i)} A_g}, \tag{8}$$

where $G(i)$ is the set of grid cells whose nearest barn is i and that are within 100 miles of it, and A_g is the number of pasture pixels in cell g. We use four windows that do not overlap: SPEI-90 at the sale date and lagged 90, 180, and 270 days. They measure drought 0–3, 3–6, 6–9, and 9–12 months before the sale and enter the model together.

Figure 1 shows the geographic distribution and reporting periods of the auction barns, together with the counties represented by pasture cells assigned to a barn within 100 miles.

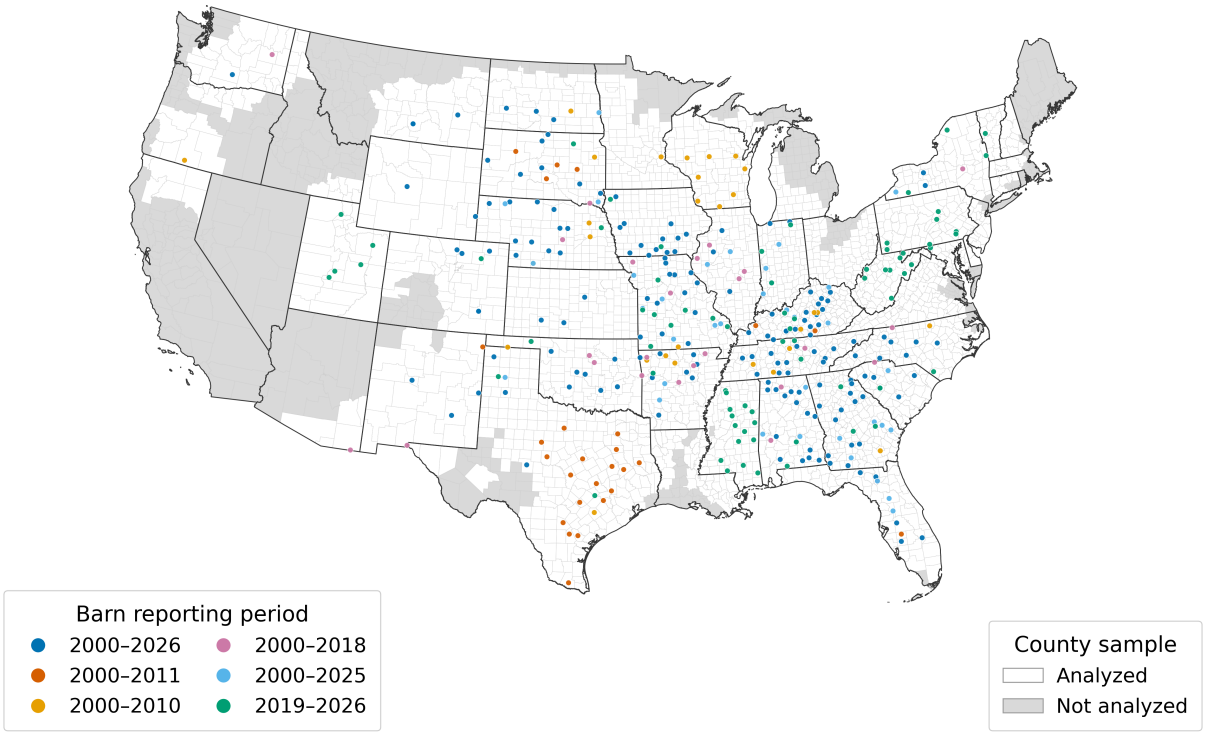


Figure 1. Geographic coverage and reporting periods of cattle auction barns

Note: Points show physical auction markets used in the analysis. Colors distinguish their reporting periods. White counties contain pasture cells assigned to the nearest eligible barn within 100 miles. Gray counties contain no assigned pasture cells and do not enter the analysis.

Our main price results focus on Medium and Large 1 feeder steers and heifers weighing 500–600 lb and Boner 80–85% slaughter cows. We also analyze prices for feeder steers and heifers of the same grade in the 400–500, 600–700, and 700–800 lb weight ranges. This captures the calf and feeder cattle market in our analysis which have different seasonal patterns. The feeder price outcomes hold weight and grade fixed, while the Boner cow outcome holds grade fixed. For number of head, we use all feeder steers, all feeder heifers, and all slaughter cows; for weight, we use all feeder steers and all feeder heifers. Head outcomes are conditional on the category appearing in the report. Separately by sex, we also use the shares of Medium and Large 1 feeder head in three weight categories: under 500, 500–600, and 600 lb or more. Number of head and weight use all feeder weights so that calves sold earlier and lighter are counted.

Table 2. Summary Statistics for Livestock Auction Outcomes and Drought Exposure

Variable	Unit	Observations	Mean	Std. dev.	D2+ (%)	D3+ (%)
Panel A: Reported number of head sold						
Slaughter cows, all grades	head	121,592	68.59	87.65		
Feeder steers, all weights and grades	head	201,031	470.11	784.87		
Feeder heifers, all weights and grades	head	202,740	378.09	537.38		
Panel B: Average weight						
Average weight: feeder steers	lb	201,027	575.03	116.16		
Average weight: feeder heifers	lb	202,740	537.76	91.92		
Panel C: Price per cwt						
Slaughter cows, Boner 80–85%	$/cwt	118,728	78.11	36.97		
Feeder steers, M&L 1, 400–500 lb	$/cwt	114,660	194.31	100.87		
Feeder steers, M&L 1, 500–600 lb	$/cwt	119,639	177.24	89.33		
Feeder steers, M&L 1, 600–700 lb	$/cwt	105,899	164.17	80.45		
Feeder steers, M&L 1, 700–800 lb	$/cwt	91,768	151.70	71.19		
Feeder heifers, M&L 1, 400–500 lb	$/cwt	123,983	171.16	89.00		
Feeder heifers, M&L 1, 500–600 lb	$/cwt	126,028	158.45	78.79		
Feeder heifers, M&L 1, 600–700 lb	$/cwt	108,472	148.92	71.73		
Feeder heifers, M&L 1, 700–800 lb	$/cwt	84,653	139.98	64.97		
Panel D: Drought exposure						
SPEI-90, 0–3 months before sale	SPEI	213,634	-0.03	0.97	10.75	6.29
SPEI-90, 3–6 months before sale	SPEI	213,634	-0.04	0.97	10.78	6.25
SPEI-90, 6–9 months before sale	SPEI	213,634	-0.01	0.95	9.80	5.52
SPEI-90, 9–12 months before sale	SPEI	213,634	0.01	0.95	9.47	5.37

Note: M&L 1 stands for Medium and Large frame, muscle grade 1. Lower SPEI values indicate drier conditions. D2+ denotes the percentage of observations with SPEI-90 ≤ -1.3, while D3+ denotes the percentage with SPEI-90 ≤ -1.6.

Table 2 presents summary statistics for the variables used in the analysis. Conditional on the category appearing in the auction report, feeder steers have the largest reported number of head, averaging 470 head per sale. Feeder heifers average 378 head per sale, while slaughter cows average approximately 69 head. In addition, feeder steers are sold at a higher average weight than feeder heifers: 575 lb compared with approximately 538 lb.

Summary statistics for prices show Boner 80–85% slaughter cows averaging $78.11/cwt. For feeder steers, average prices decline across the reported weight categories, from $194.31/cwt for cattle weighing 400–500 lb to $177.24, $164.17, and $151.70/cwt for the 500–600, 600–700, and 700–800 lb categories, respectively. We observe the same pattern for feeder heifers, with

average prices declining from $171.16/cwt in the 400–500 lb category to $158.45, $148.92, and $139.98/cwt in the three successively heavier categories.

Finally, mean SPEI-90 is close to zero in all four windows before the sale, indicating typical conditions on average. However, standard deviations between 0.95 and 0.97 show substantial variation around those means. Severe-or-worse drought conditions (D2 or D3, as defined in table 1) occur in 10.75% of barn-sale observations in the 0–3 month window and 10.78% in the 3–6 month window. The corresponding shares are 9.80% for the 6–9 month window and 9.47% for the 9–12 month window. Extreme-or-worse drought conditions (D3 or worse) occur in approximately 5.4–6.3% of barn-sale observations, depending on the window.

Empirical Model

We estimate the effects of drought during the year before the sale using a fixed effect model. For outcomes estimated with OLS,

$$y_{is} = \alpha_{im} + \gamma_w + \sum_{k \in K} \beta_k \, Drought_{iks} + \sum_{k \in K} \delta_k \, (Drought_{iks})^2 + \epsilon_{is}, \tag{9}$$

where y_{is} is the log of price or weight, or a weight-category share in levels; s denotes sale; w denotes the calendar year-week of the sale; i indicates barn; m indicates month of the year; and k indicates the drought window. α_{im} is the barn-month fixed effect that controls the usual seasonal conditions of barn i in month m, and γ_w refers to year-week FE, controlling for shocks that affect all barns in a given week (including futures, feed costs from tariffs, etc.).[1] ϵ_{is} denotes the standard errors clustered by barn. $Drought_{iks}$ is the pasture-weighted SPEI-90 measure defined in Equation (8) for barn i, sale s, and window k. It is measured in raw SPEI units: lower values indicate drier conditions, while zero indicates typical conditions for that place and time of year. The four windows

do not overlap and enter together,

$$K = \{0\text{–}3,\ 3\text{–}6,\ 6\text{–}9,\ 9\text{–}12 \text{ months}\},$$

so each pair of coefficients measures the effect of drought in that quarter holding drought in the other quarters fixed. We use both linear and squared terms to capture non-linearity in the effects of droughts. A causal interpretation requires local drought to be conditionally exogenous after accounting for the fixed effects.

Prices and weight are estimated in logs with OLS, and weight-category shares are estimated in levels with OLS. Number of head is estimated with Poisson,

$$\mu_{is} \equiv E[y_{is} \mid X] = \exp\left(\alpha_{im} + \gamma_w + \sum_{k \in K} \beta_k Drought_{iks} + \sum_{k \in K} \delta_k Drought_{iks}^2\right). \tag{10}$$

All regressions are unweighted, and the drought variable enters in raw SPEI units.

Because drought enters quadratically, its marginal effect depends on the level of drought. The marginal effect of drought is:

$$ME_k^{Drought} = 100\,(\beta_k + 2\delta_k Drought)\,. \tag{11}$$

Although this marginal effect describes the slope of the response at a given drought level, we are primarily interested in how severe and extreme droughts affect livestock sales relative to normal conditions. These comparisons are more policy relevant and easier to interpret. Therefore, the tables report the discrete effect of moving from typical conditions ($Drought = 0$) to a specified drought level:

$$\Delta_k(Drought) = 100\left[\exp\left(\beta_k Drought + \delta_k Drought^2\right) - 1\right]. \tag{12}$$

Our main results evaluate these effects at severe drought, D2 ($Drought = -1.3$), while extreme

drought (D3, $Drought = -1.6$) are presented in the appendix.

Results

We report the effects of droughts relative to typical conditions. We focus the text and main tables on severe droughts (D2 in the Drought Monitor). The corresponding extreme drought (D3) tables are in the appendix. In each table, the four quarterly windows enter together, so each estimate holds drought in the other quarters fixed. The cumulative row sets all four quarterly windows to the indicated drought level simultaneously.

Reported Number of Head Sold

Table 3 reports the estimated effects on the number of head. We found that a severe drought around the barn 0 to 3 months before the sale date increases the reported number of cow head by 9.6%. The amount remains higher by 4.6% at 3–6 months, 3.0% at 6–9 months, and 4.2% at 9–12 months. Severe drought throughout the year increases the reported number of cows by 22.9%. These results align with Ritten et al. (2010), who model partial liquidation as a principal strategy producers can use during drought, while Wilmer et al. (2016) document that drought affects ranch planning and practice; they argue that most part of the farmers use destocking as a primary response during a drought. In particular, they show that 80% of the cattle were sold in an Oklahoma county. In general, authors found that severe drought is responsible for reduced forage availability, consequently increasing the cost of feed, and, by the end, putting pressure on producers to liquidate the herd instead of maintaining it (Countryman, Paarlberg, and Lee 2016; Rodziewicz, Dice, and Cowley 2023; Beck et al. 2024; Bittmann et al. 2025). Our results complement their findings by documenting increased cow marketing at the auction level.

Table 3. Effect of Severe Drought (D2) on Reported Number of Head Sold by Cattle Category (%)

Drought (D2) window before sale	All-grade cows	All-weight steers	All-weight heifers
0-3 months	9.55*** (1.28)	2.74*** (0.66)	4.39*** (0.66)
3-6 months	4.56*** (1.00)	0.94 (0.71)	1.92*** (0.65)
6-9 months	3.02*** (0.95)	1.46** (0.58)	2.66*** (0.56)
9-12 months	4.17*** (1.02)	−0.24 (0.50)	0.79 (0.52)
Cumulative 0–12 months	22.92*** (3.83)	4.97*** (1.73)	10.09*** (1.90)
Barn × month-of-year FE	Yes	Yes	Yes
Calendar year × ISO-week FE	Yes	Yes	Yes
Observations	121,592	201,031	202,740

Note: Estimates compare severe drought (D2, SPEI = −1.3) with normal conditions (SPEI = 0), holding the other drought windows fixed. Effects are percentages. Barn-clustered standard errors are in parentheses. The cumulative row sets all four windows to D2 simultaneously. Its standard error is calculated using the delta method. *** p<0.01, ** p<0.05, * p<0.10.

The feeder-cattle results also suggest that drought affects marketing and retention decisions differently across cattle categories. During severe drought across all four quarterly windows, reported heifer head increases by 10.09%, about twice the estimated 4.97% increase for steers. Heifer marketings respond significantly to drought in the first three windows, whereas the steer response is statistically significant only in the 0–3 and 6–9 month windows. The larger heifer point estimate is consistent with reduced retention of potential breeding replacements and accelerated sales of animals already intended for the feeder market. In a case study, Whelan et al. (2013) argue that a trade-off exists between immediate selling for financial relief and future productive capacity, finding that selling 10% of the herd yields the best cash-flow outcome. Also, Beck

et al. (2024) report reduced fertility among replacement heifers born to cows exposed to nutritional and heat stress during pregnancy, which could explain high rates of heifers being sold during drought periods. This interpretation aligns with Countryman, Paarlberg, and Lee (2016), whose dynamic model links drought-induced changes in feed costs and expected returns to breeding-herd adjustments and subsequent calf supply.

Weight

Table 4 shows the effect of drought on the weight of steers and heifers. Overall, we found that a severe drought within a year from the sale date reduces the weight of feeder calves. Severe drought in the 0–3 months before sale reduces average steer weight by 1.16% and average heifer weight by 0.91%. The corresponding reductions for drought in the 3–6-month window are 1.24% for steers and 1.00% for heifers. Negative estimates persist across all drought windows. The 12-month cumulative window indicates reductions of 3.80% for steers and 2.57% for heifers. In a Wyoming study, Scasta et al. (2015) found that each 1-inch reduction in precipitation lowers calf weaning weight by 7 to 14 lb and average daily gain by 0.03 to 0.07 lb. Another important point is that Beck et al. (2024) studied drought-related destocking and found that limited forage can cause calves to bypass the stocker phase and enter feedlots younger and at lower body weights. Our auction-barn results align with the findings in these papers.

Table 4. Effect of Severe Drought (D2) on Average Weight of Feeder Steers and Heifers (%)

Drought (D2) window before sale	All-weight steers	All-weight heifers
0-3 months	−1.16*** (0.10)	−0.91*** (0.08)
3-6 months	−1.24*** (0.13)	−1.00*** (0.10)
6-9 months	−0.95*** (0.12)	−0.49*** (0.10)
9-12 months	−0.51*** (0.10)	−0.19** (0.08)
Cumulative 0–12 months	−3.80*** (0.35)	−2.57*** (0.29)
Barn × month-of-year FE	Yes	Yes
Calendar year × ISO-week FE	Yes	Yes
Observations	201,027	202,740

Note: Estimates compare severe drought (D2, SPEI = −1.3) with normal conditions (SPEI = 0), holding the other drought windows fixed. Effects are percentages. Barn-clustered standard errors are in parentheses. The cumulative row sets all four windows to D2 simultaneously. Its standard error is calculated using the delta method. *** p<0.01, ** p<0.05, * p<0.10.

The weight-distribution results in Table 5 reinforce the evidence that lighter cattle are entering the market. Severe drought during the 0–3-month window increases the share weighing less than 500 lb by 1.21 percentage points for steers and 1.24 percentage points for heifers. The estimated increase reaches 1.65 percentage points for both sexes in the 3–6 month window. Under severe drought in all four windows (the cumulative 0-12 months row), the share below 500 lb increases by 4.07 percentage points for steers and 3.87 percentage points for heifers. These changes are accompanied by declines in the combined shares of heavier cattle. The composition results provide additional evidence consistent with the marketing adjustments discussed by Mulliniks, Waterman, and Geary (2013), including the possibility of selling lighter calves when maintaining them becomes costly.

Table 5. Effect of Severe Drought (D2) on Shares of Head Sold by Sex and Weight Bracket (Percentage Points)

Drought (D2) window before sale	Steers			Heifers		
	Under 500 lb	500–600 lb	600 lb and over	Under 500 lb	500–600 lb	600 lb and over
0-3 months	1.21*** (0.18)	−0.43*** (0.14)	−0.78*** (0.23)	1.24*** (0.18)	−0.87*** (0.13)	−0.37 (0.22)
3-6 months	1.65*** (0.24)	0.10 (0.13)	−1.75*** (0.28)	1.65*** (0.23)	−0.43*** (0.15)	−1.22*** (0.23)
6-9 months	1.02*** (0.24)	−0.20 (0.14)	−0.82*** (0.26)	0.84*** (0.23)	−0.44*** (0.14)	−0.40* (0.22)
9-12 months	0.19 (0.18)	−0.58*** (0.13)	0.38* (0.22)	0.14 (0.20)	−0.69*** (0.14)	0.55*** (0.19)
Cumulative 0–12 months	4.07*** (0.67)	−1.11*** (0.33)	−2.96*** (0.73)	3.87*** (0.65)	−2.44*** (0.36)	−1.44** (0.64)
Outcome mean (%)	27.05	25.11	47.84	35.28	27.94	36.79
Barn × month-of-year FE	Yes	Yes	Yes	Yes	Yes	Yes
Calendar year × ISO-week FE	Yes	Yes	Yes	Yes	Yes	Yes
Observations	110,976	110,976	110,976	118,370	118,370	118,370

Note: Outcomes are shares of Medium and Large 1 feeder head in three exhaustive weight brackets, estimated separately by sex. Sales with fewer than 20 head of that sex are excluded. A bracket's share is zero when no head are recorded in that bracket. Estimates compare severe drought (D2, SPEI = −1.3) with normal conditions (SPEI = 0), holding the other drought windows fixed. Effects are percentage points. Barn-clustered standard errors are in parentheses. The cumulative row sets all four windows to D2 simultaneously. Its standard error is calculated using the delta method. *** p<0.01, ** p<0.05, * p<0.10.

The results also reveal some differences between steers and heifers. The increase in the under-500-pound category is similar for the two sexes, but the middle-weight response is more consistently negative for heifers. This suggests that drought may alter marketing decisions differently depending on whether animals are steers or potential replacement females.

Given more units are being sold, and lighter than usual, what is the effect on prices? That is what we answer in the next subsection.

Prices

Table 6 shows an immediate price discount after severe drought. Severe drought during the three months before sale reduces boner-cow prices by 0.44%, while prices for 500–600 lb steers and heifers decline by 0.40% and 0.27%, respectively. Our barn-level estimates show that drought can initially lower local auction prices as additional cattle enter the market. Evidence shows that drought increases liquidation, shifting short-run supply and, consequently, decreasing cattle prices in local areas close to the environmental issue (Leister, Lee, and Paarlberg 2013; Leister, Paarlberg, and Lee 2015; Countryman, Paarlberg, and Lee 2016; Fleming-Muñoz, Whitten, and Bonnett 2023). Our estimates, however, capture local price differences after accounting for shocks common to all barns in a given year-week, rather than the total national price response to drought, as in those studies.

The price estimates associated with more distant drought exposure (three months or more before sale) are generally smaller and sometimes positive. For cows, drought during the 3–6 month window produces a 0.16% discount, significant at the 10% level, while the 9–12 month estimate is a positive 0.18%. For 500–600 lb steers, the 3–6 month estimate is a positive 0.10%, and the remaining estimates are not statistically significant at the 5% level. None of the later-window estimates for similarly weighted heifers are statistically significant. The joint four-window estimates are also not statistically different from zero for the three principal categories. These findings indicate limited evidence of a persistent local price discount within these categories.

Table 6. Effect of Severe Drought (D2) on Price per cwt for Principal Cattle Categories (%)

Drought (D2) window before sale	Boner cows	Steers 500–600	Heifers 500–600
0-3 months	−0.44*** (0.09)	−0.40*** (0.06)	−0.27*** (0.06)
3-6 months	−0.16* (0.10)	0.10** (0.05)	0.05 (0.05)
6-9 months	0.08 (0.10)	0.07 (0.06)	0.02 (0.06)
9-12 months	0.18** (0.09)	0.04 (0.05)	0.02 (0.05)
Cumulative 0–12 months	−0.34 (0.30)	−0.18 (0.14)	−0.18 (0.15)
Barn × month-of-year FE	Yes	Yes	Yes
Calendar year × ISO-week FE	Yes	Yes	Yes
Observations	118,728	119,639	126,028

Note: Estimates compare severe drought (D2, SPEI = −1.3) with normal conditions (SPEI = 0), holding the other drought windows fixed. Effects are percentages. Barn-clustered standard errors are in parentheses. The cumulative row sets all four windows to D2 simultaneously. Its standard error is calculated using the delta method. *** p<0.01, ** p<0.05, * p<0.10.

Tables 7 and 8, together with Figure 2, reveal that the immediate price discount is concentrated among the lightest cattle. For 400–500 lb cattle, severe drought during the 0–3 month window reduces steer prices by 0.59% and heifer prices by 0.37%. The discounts become smaller at 500–600 lb and are close to zero for 700–800 lb cattle. One possible explanation is that drought simultaneously increases the supply of lighter calves and reduces demand from buyers who depend on pasture to add weight. This interpretation is consistent with the broader finding of Dhuyvetter and Schroeder (2000) that feeder-cattle price–weight relationships depend on feed prices and expected fed-cattle prices. The present results therefore suggest that drought-related price adjustments differ across stages of feeder development.

Table 7. Effect of Severe Drought (D2) on Feeder-Steer Price per cwt by Average-Weight Bracket (%)

Drought (D2) window before sale	400–500 lb	500–600 lb	600–700 lb	700–800 lb
0-3 months	−0.59*** (0.07)	−0.40*** (0.06)	−0.11** (0.05)	−0.02 (0.04)
3-6 months	0.07 (0.06)	0.10** (0.05)	0.20*** (0.04)	0.22*** (0.04)
6-9 months	0.11* (0.06)	0.07 (0.06)	0.07 (0.05)	0.05 (0.05)
9-12 months	−0.05 (0.06)	0.04 (0.05)	0.11*** (0.04)	0.11*** (0.04)
Cumulative 0–12 months	−0.47*** (0.16)	−0.18 (0.14)	0.27** (0.12)	0.36*** (0.12)
Barn × month-of-year FE	Yes	Yes	Yes	Yes
Calendar year × ISO-week FE	Yes	Yes	Yes	Yes
Observations	114,660	119,639	105,899	91,768

Note: Estimates compare severe drought (D2, SPEI = −1.3) with normal conditions (SPEI = 0), holding the other drought windows fixed. Effects are percentages. Barn-clustered standard errors are in parentheses. The cumulative row sets all four windows to D2 simultaneously. Its standard error is calculated using the delta method. *** p<0.01, ** p<0.05, * p<0.10.

Table 8. Effect of Severe Drought (D2) on Feeder-Heifer Price per cwt by Average-Weight Bracket (%)

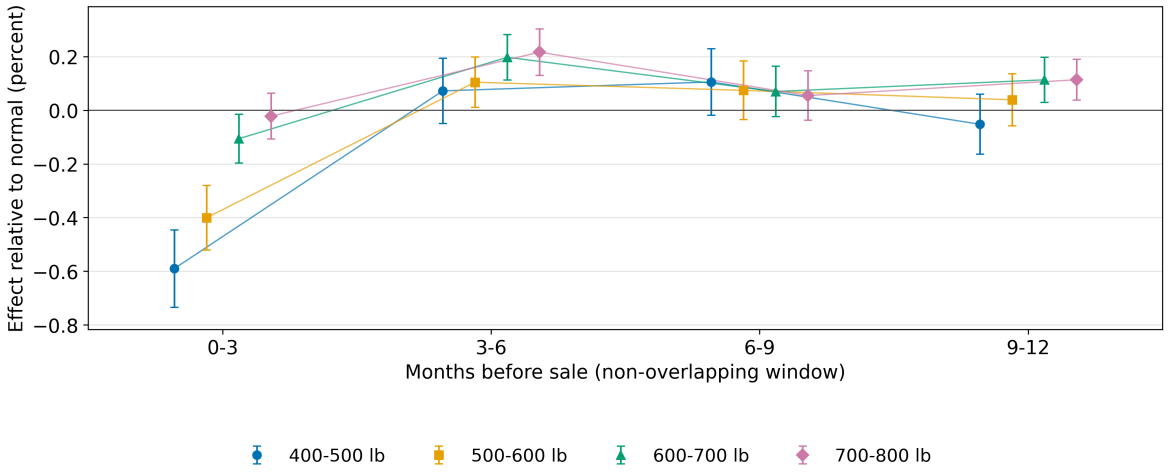
Drought (D2) window before sale	400–500 lb	500–600 lb	600–700 lb	700–800 lb
0-3 months	−0.37*** (0.07)	−0.27*** (0.06)	−0.08 (0.05)	0.03 (0.06)
3-6 months	0.10 (0.06)	0.05 (0.05)	0.12** (0.05)	0.23*** (0.05)
6-9 months	0.11* (0.06)	0.02 (0.06)	0.03 (0.06)	0.08 (0.05)
9-12 months	0.00 (0.06)	0.02 (0.05)	0.03 (0.05)	0.20*** (0.07)
Cumulative 0–12 months	−0.17 (0.18)	−0.18 (0.15)	0.11 (0.15)	0.54*** (0.15)
Barn × month-of-year FE	Yes	Yes	Yes	Yes
Calendar year × ISO-week FE	Yes	Yes	Yes	Yes
Observations	123,983	126,028	108,472	84,653

Note: Estimates compare severe drought (D2, SPEI = −1.3) with normal conditions (SPEI = 0), holding the other drought windows fixed. Effects are percentages. Barn-clustered standard errors are in parentheses. The cumulative row sets all four windows to D2 simultaneously. Its standard error is calculated using the delta method. *** p<0.01, ** p<0.05, * p<0.10.

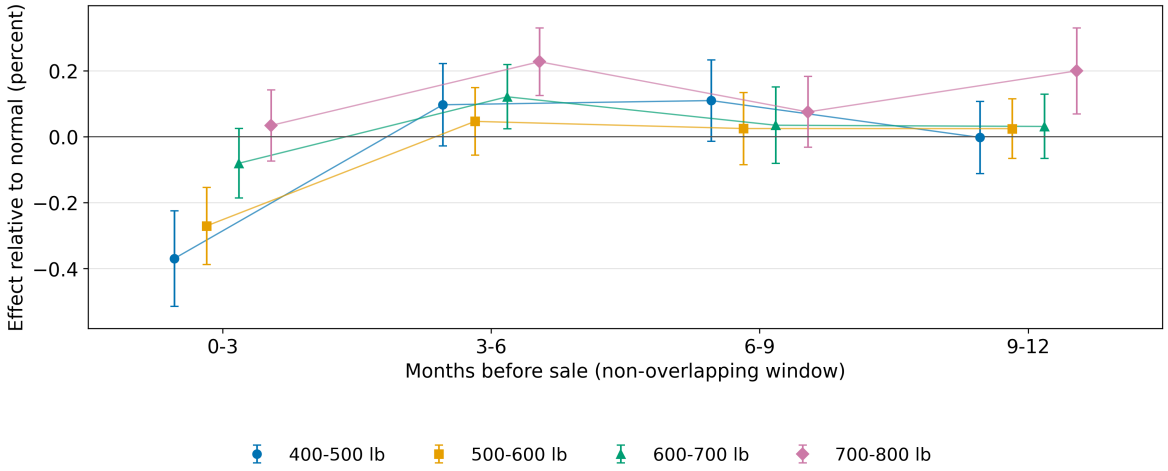
When we look at the cumulative response in a year, we find positive cumulative coefficients for heavier steers (0.27% for 600–700 pounds and 0.36% for 700–800 pounds) and for 700–800-pound heifers (0.54%) and negative cumulative coefficients for lighter steers (-0.47% for 400–500 pounds). Therefore, drought fundamentally reshapes feeder cattle price-weight relationships because lighter calves and heavier backgrounded cattle respond differently to pasture shortages and rising grain expenses. Feeder cattle price slides reflect the cost of adding animal gain, where price per hundredweight decreases as animal weight rises (Schulz, Boetel, and Dhuyvetter 2018; Hudson and Dennis 2025; Dhuyvetter, Schroeder, and Prevatt 2001). Martinez, Boyer, and Burdine (2021), in a study for Tennessee, found that rising corn prices negatively affect lighter calves, while heavier

cattle prices can gain a relative advantage as feedlots seek animals requiring less yard feeding. Our results are consistent with those findings.

Together, our results for prices indicate that price discounts are concentrated in the first three months after drought, especially for the lightest categories, while positive effects appear at longer lags, especially for heavier feeder cattle. However, compared with the results for number of head, the impacts of drought on prices are smaller in percentage terms.



(a) Effect of severe drought (D2) on price per cwt: Medium and Large 1 feeder steers



(b) Effect of severe drought (D2) on price per cwt: Medium and Large 1 feeder heifers

Figure 2. Effect of severe drought (D2) on feeder-cattle price per cwt by average-weight bracket and drought window

Note: Points compare D2 exposure with normal conditions and show 95% confidence intervals clustered by barn. Each panel uses one common y-axis range across its four brackets.

Conclusions

This article analyzes how drought conditions in pastures surrounding cattle auction barns affect the reported number of head sold, weights, and prices for feeder cattle and cull cows. Our results indicate that severe drought during the three months before an auction significantly increases the number of head sold in all three cattle categories, with the largest effect for cull cows.

We also find that steers and heifers are sold at lighter weights following drought, consistent with drought affecting weaning and marketing decisions. Finally, prices for the main categories of cows, steers, and heifers decrease following drought, although these discounts are concentrated when drought occurs during the three months before the sale. Among feeder cattle, price discounts are larger for lighter categories and concentrated in the first three months after drought, while positive effects appear at longer lags for heavier categories. These differences suggest that drought affects market conditions differently across feeder-cattle weight categories.

Overall, our results show that drought can have important effects on local livestock markets by increasing short-run cattle marketings, shifting feeder cattle toward lighter weights, and modestly reducing prices. Our analysis focuses on effects during the first year following drought conditions. Future research should examine how these effects evolve beyond the first year and interact with the cattle cycle.

Our results have important implications for current efforts to increase U.S. cattle and beef supplies. Policies aimed at rebuilding the cattle herd ultimately depend on individual producers retaining breeding animals rather than marketing them. We find that drought works against this objective by increasing cow and heifer marketings as producers adjust to deteriorating forage conditions. These additional marketings increase cattle supplies in the short run but can reduce the breeding inventory available to produce cattle in subsequent years. This distinction is important because policies that increase beef supplies in the short run are not necessarily consistent with the producer decisions required to expand domestic cattle supplies over the longer run. More broadly, our results show that production conditions such as drought can affect the timing and magnitude of

herd expansion even when cattle prices provide strong incentives for producers to retain breeding stock.

Our results also show that the effects of drought are more apparent in cattle marketing decisions than in local cattle prices. We find relatively large increases in the number of cows and heifers marketed during drought, while the estimated price effects are comparatively small and concentrated in the months immediately following drought. Thus, cattle prices alone may provide an incomplete measure of how producers are responding to drought. Even when national cattle prices remain high, local drought conditions can lead producers to market breeding animals that might otherwise have been retained.

Footnotes

[1]Note the difference between the two time indices. m is the month of the year only, so February of 2011 and February of 2019 are the same m and the barn-month fixed effect α_{im} captures the normal February of barn i in any year. w is a specific week of a specific year, so the week fixed effect γ_w is different for every week in the sample, and it captures what happened to all barns in that particular week.

AI Assistance

AI tools (OpenAI's ChatGPT/Codex) were used to assist with language editing and the development of R code. The authors take full responsibility for the manuscript, code, and reported results.

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Appendix: Extreme Drought Results

The following tables repeat the family comparisons at D3 (SPEI = −1.6) relative to normal conditions.

Table A1. Effect of Extreme Drought (D3) on Reported Number of Head Sold by Cattle Category (%)

Drought (D3) window before sale	All-grade cows	All-weight steers	All-weight heifers
0-3 months	12.60*** (1.83)	3.66*** (0.90)	5.88*** (0.89)
3-6 months	6.10*** (1.39)	1.25 (0.96)	2.48*** (0.89)
6-9 months	3.99*** (1.35)	2.13*** (0.80)	3.57*** (0.78)
9-12 months	5.71*** (1.40)	−0.41 (0.67)	0.90 (0.69)
Cumulative 0–12 months	31.33*** (5.67)	6.75*** (2.39)	13.39*** (2.61)
Barn × month-of-year FE	Yes	Yes	Yes
Calendar year × ISO-week FE	Yes	Yes	Yes
Observations	121,592	201,031	202,740

Note: Estimates compare extreme drought (D3, SPEI = −1.6) with normal conditions (SPEI = 0), holding the other drought windows fixed. Effects are percentages. Barn-clustered standard errors are in parentheses. The cumulative row sets all four windows to D3 simultaneously. Its standard error is calculated using the delta method. *** p<0.01, ** p<0.05, * p<0.10.

Table A2. Effect of Extreme Drought (D3) on Average Weight of Feeder Steers and Heifers (%)

Drought (D3) window before sale	All-weight steers	All-weight heifers
0-3 months	−1.59*** (0.14)	−1.25*** (0.11)
3-6 months	−1.68*** (0.17)	−1.37*** (0.14)
6-9 months	−1.34*** (0.16)	−0.73*** (0.13)
9-12 months	−0.77*** (0.14)	−0.34*** (0.10)
Cumulative 0–12 months	−5.27*** (0.47)	−3.64*** (0.38)
Barn × month-of-year FE	Yes	Yes
Calendar year × ISO-week FE	Yes	Yes
Observations	201,027	202,740

Note: Estimates compare extreme drought (D3, SPEI = −1.6) with normal conditions (SPEI = 0), holding the other drought windows fixed. Effects are percentages. Barn-clustered standard errors are in parentheses. The cumulative row sets all four windows to D3 simultaneously. Its standard error is calculated using the delta method. *** p<0.01, ** p<0.05, * p<0.10.

Table A3. Effect of Extreme Drought (D3) on Price per cwt for Principal Cattle Categories (%)

Drought (D3) window before sale	Boner cows	Steers 500–600	Heifers 500–600
0-3 months	−0.57*** (0.12)	−0.53*** (0.08)	−0.37*** (0.08)
3-6 months	−0.22* (0.13)	0.13** (0.07)	0.07 (0.07)
6-9 months	0.07 (0.13)	0.09 (0.08)	0.04 (0.08)
9-12 months	0.17 (0.12)	0.06 (0.07)	0.04 (0.06)
Cumulative 0–12 months	−0.55 (0.40)	−0.25 (0.19)	−0.22 (0.20)
Barn × month-of-year FE	Yes	Yes	Yes
Calendar year × ISO-week FE	Yes	Yes	Yes
Observations	118,728	119,639	126,028

Note: Estimates compare extreme drought (D3, SPEI = −1.6) with normal conditions (SPEI = 0), holding the other drought windows fixed. Effects are percentages. Barn-clustered standard errors are in parentheses. The cumulative row sets all four windows to D3 simultaneously. Its standard error is calculated using the delta method. *** p<0.01, ** p<0.05, * p<0.10.

Table A4. Effect of Extreme Drought (D3) on Feeder-Steer Price per cwt by Average-Weight Bracket (%)

Drought (D3) window before sale	400–500 lb	500–600 lb	600–700 lb	700–800 lb
0-3 months	−0.79*** (0.10)	−0.53*** (0.08)	−0.14** (0.06)	−0.02 (0.06)
3-6 months	0.07 (0.08)	0.13** (0.07)	0.27*** (0.06)	0.31*** (0.06)
6-9 months	0.14 (0.09)	0.09 (0.08)	0.10 (0.07)	0.09 (0.06)
9-12 months	−0.05 (0.08)	0.06 (0.07)	0.16*** (0.06)	0.16*** (0.05)
Cumulative 0–12 months	−0.64*** (0.21)	−0.25 (0.19)	0.39** (0.16)	0.54*** (0.16)
Barn × month-of-year FE	Yes	Yes	Yes	Yes
Calendar year × ISO-week FE	Yes	Yes	Yes	Yes
Observations	114,660	119,639	105,899	91,768

Note: Estimates compare extreme drought (D3, SPEI = −1.6) with normal conditions (SPEI = 0), holding the other drought windows fixed. Effects are percentages. Barn-clustered standard errors are in parentheses. The cumulative row sets all four windows to D3 simultaneously. Its standard error is calculated using the delta method. *** p<0.01, ** p<0.05, * p<0.10.

Table A5. Effect of Extreme Drought (D3) on Feeder-Heifer Price per cwt by Average-Weight Bracket (%)

Drought (D3) window before sale	400–500 lb	500–600 lb	600–700 lb	700–800 lb
0-3 months	−0.50*** (0.10)	−0.37*** (0.08)	−0.12 (0.07)	0.04 (0.07)
3-6 months	0.11 (0.09)	0.07 (0.07)	0.18*** (0.07)	0.32*** (0.07)
6-9 months	0.15* (0.09)	0.04 (0.08)	0.06 (0.08)	0.11 (0.07)
9-12 months	0.02 (0.08)	0.04 (0.06)	0.04 (0.07)	0.25*** (0.09)
Cumulative 0–12 months	−0.22 (0.24)	−0.22 (0.20)	0.16 (0.20)	0.72*** (0.20)
Barn × month-of-year FE	Yes	Yes	Yes	Yes
Calendar year × ISO-week FE	Yes	Yes	Yes	Yes
Observations	123,983	126,028	108,472	84,653

Note: Estimates compare extreme drought (D3, SPEI = −1.6) with normal conditions (SPEI = 0), holding the other drought windows fixed. Effects are percentages. Barn-clustered standard errors are in parentheses. The cumulative row sets all four windows to D3 simultaneously. Its standard error is calculated using the delta method. *** p<0.01, ** p<0.05, * p<0.10.